



HORMONES, SEXUAL HEALTH & MISC

PT-141

PT-141 – 10 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	Bremelanotide; Ac-Nle-cyclo(-Asp-His-D-Phe-Arg-Trp-Lys-)-OH
SYNONYMS / ALIASES	PT-141; bremelanotide; melanocortin receptor agonist
CAS NUMBER	189691-06-3
MOLECULAR FORMULA	C50H68N14O10
MOLECULAR WEIGHT	1025.18 g/mol
AMINO ACID SEQUENCE	Ac-Nle-cyclo[Asp-His-D-Phe-Arg-Trp-Lys]-OH (7 residues; N-terminal acetyl-norleucine; lactam-bridged cyclic hexapeptide via Asp-Lys side chains; free C-terminal carboxylate on Lys)
SOURCE	Synthetic (solid-phase peptide synthesis with lactam cyclization)

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA)
CONTAINER	10 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 28 days in bacteriostatic water
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Lactam-bridge cyclization confers significant enzymatic stability. The tryptophan residue is light-sensitive; protect from prolonged light exposure.

STRUCTURAL NOTES

PT-141 (bremelanotide) is a synthetic 7-amino-acid cyclic peptide derived from the alpha-MSH agonist melanotan-II by removal of the C-terminal amide group (giving a free C-terminal carboxylate). It is structurally a lactam-bridged cyclic peptide: the side-chain carboxylate of aspartate forms an amide bond with the epsilon-amino group of lysine, creating a constrained ring. The N-terminus is acetylated and bears a non-natural norleucine (Nle, an isostere of methionine without the oxidation-prone sulfur). It is classified as a non-selective melanocortin receptor agonist with activity at MC1R, MC3R, MC4R, and MC5R.

SELECTED PEER-REVIEWED REFERENCES

1. Molinoff P.B. et al. - *PT-141: a melanocortin agonist for the treatment of sexual dysfunction* - Annals of the New York Academy of Sciences, 2003 - DOI: 10.1196/annals.1290.034
2. Diamond L.E. et al. - *Double-blind, placebo-controlled evaluation of the safety, pharmacokinetic properties and pharmacodynamic effects of intranasal PT-141, a melanocortin receptor agonist, in healthy males and patients with mild-to-moderate erectile dysfunction* - International Journal of Impotence Research, 2004 - DOI: 10.1038/sj.ijir.3901168
3. Kingsberg S.A. et al. - *Bremelanotide for the treatment of hypoactive sexual desire disorder: two randomized phase 3 trials* - Obstetrics & Gynecology, 2019 - DOI: 10.1097/AOG.0000000000003500

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

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Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.